

DIFFERENTIATION PROBLEM SHEET

Complete the problems below. Solutions and marking available on request.

1. Find the derivative $\frac{dy}{dx}$ for

- (a) the constant function $y = 6$,
- (b) the line $y = 6x + 3$,
- (c) the curve $y = 2x^4 - 3x^3 + \sqrt{x} + 7/x$.

2. Consider the trigonometric functions $\sin(x)$ and $\cos(x)$.

- (a) What is the derivative of $\sin(x)$? What is the derivative of $\cos(x)$?
- (b) Let $f(x) = \sin(x)$. What is the fourth derivative $f^{(4)}(x)$?
- (c) Find $f^{(2023)}(x)$, where $f^{(2023)}$ is the 2023rd derivative of f .

This question is soooooo last year, ugh.

Hint: what is the nearest multiple of 4 to 2023? How do the derivatives of sine behave; how do they cycle?

3. Let a be some positive real number.

- (a) Given $y = a^x$, find $\frac{dy}{dx}$.
- (b) Suppose $a = e$, where e is Euler's constant. Use (a) to evaluate the derivative in this case, giving your answer in its simplest form.

4. If $f(t) = e^{4t+3}$, find $f'(t)$.

5. If $y = (x^2 + 3)\sin(x)$, find $\frac{dy}{dx}$.

6. If $y = 5^t \sin(3t)$, find $\frac{dy}{dt}$.

7. A curve C is given implicitly by $2y = x^3 + \sin(y) + xe^y + 7$. Find the gradient of the curve at the point $(1, \pi)$, leaving your answer in terms of e .

8. Let $f(y) = \ln(\tan(y))$. Is the value $f'(0)$ well defined? Justify your answer.

9. Consider a curve defined by the parametric equations $x = 2t + 1$ and $y = t^2 - 1$.

- (a) Find the points where the curve crosses the coordinate axes.
- (b) Find an expression for $\frac{dy}{dx}$ in terms of x .

10. Consider a curve defined by the parametric equations $x = \tan(\theta)$ and $y = \cos(\theta)$.

- (a) Find $\frac{dy}{dx}$.
- (b) Find the equation of the normal line to the curve at $\theta = \pi/2$.